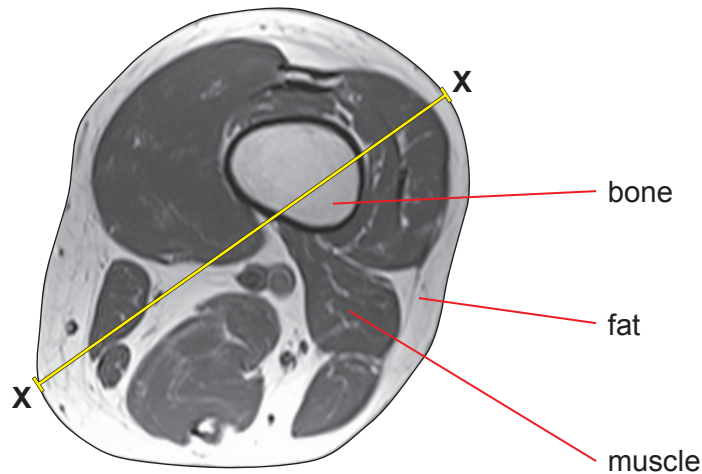


Option B: Human Musculoskeletal Anatomy

9. The image shows a transverse section through the human thigh using an imaging technique called magnetic resonance imaging (MRI). The image shows bone, fat and muscle.



- (a) (i) Name the bone shown within the thigh. [1]

- (ii) At point X-X on the image the thigh has an actual width of 150 mm. Calculate the magnification of the image. [2]

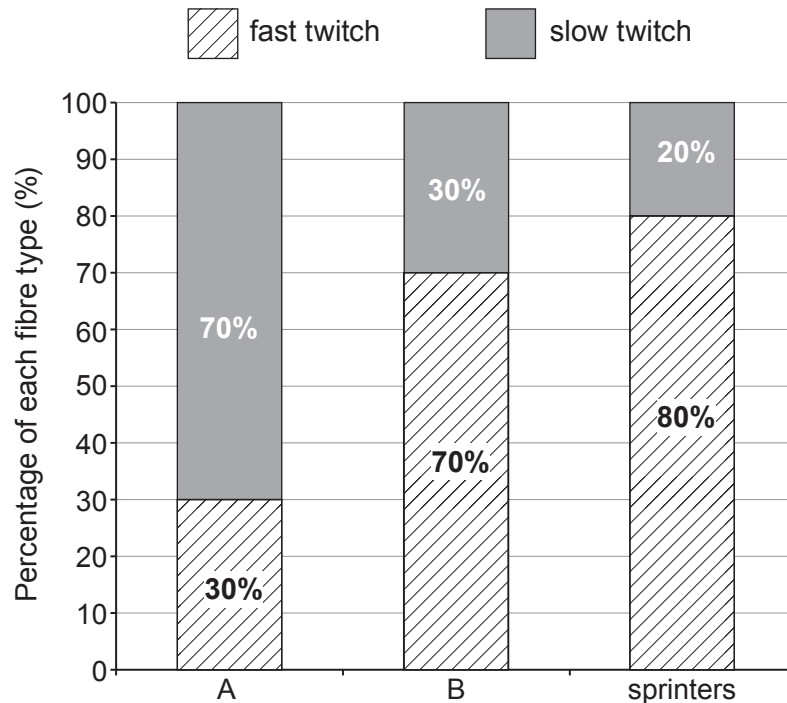
magnification =

- (iii) Bones have an outer layer of compact bone. State which inorganic components strengthen the bone. [1]

- (iv) Explain how the structure of the thigh bone enables nutrients to be supplied to the osteoblasts and osteoclasts. [2]



The proportion of fast and slow twitch muscle fibres was investigated in the muscles of the thigh in a range of athletes. The graph below shows the percentage of slow and fast twitch fibres in the muscle samples.



- (v) Investigators observed a difference in colour between fast and slow twitch fibres. Suggest what this difference may be and explain your answer. [2]

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- (vi) Explain why the percentage of fibres was calculated rather than using the total number of each fibre type. [2]

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- (vii) Explain why the athletes chosen were of the same age, gender and ethnicity. [1]

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- (viii) It was concluded that Athlete B was a weightlifter as further analysis of the tissue found high lactate levels. With reference to the graph and your own knowledge explain how they reached this conclusion. [3]

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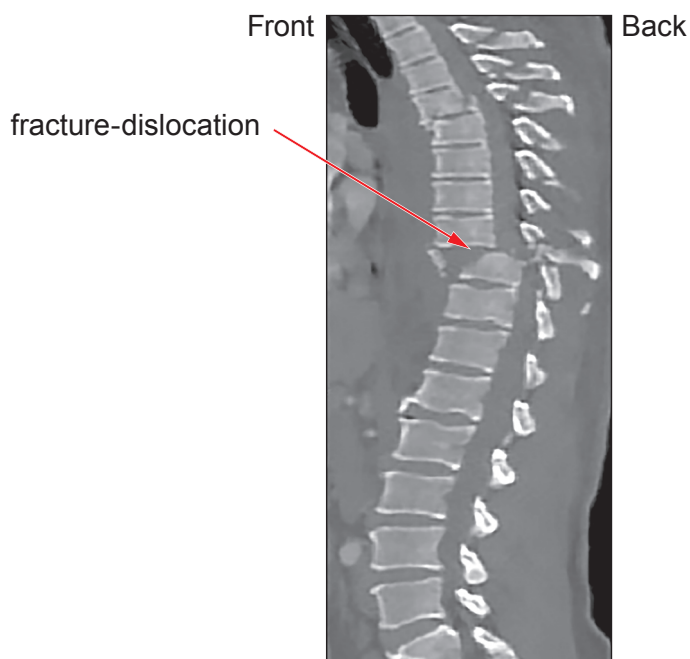
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- (b) The image below shows an X-ray of a human vertebral column showing a fracture to the axial skeleton.



- (i) State **two** functions of this part of the skeleton. [1]

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- (ii) With reference to the X-ray, state which region of the vertebral column contains a fracture. [1]

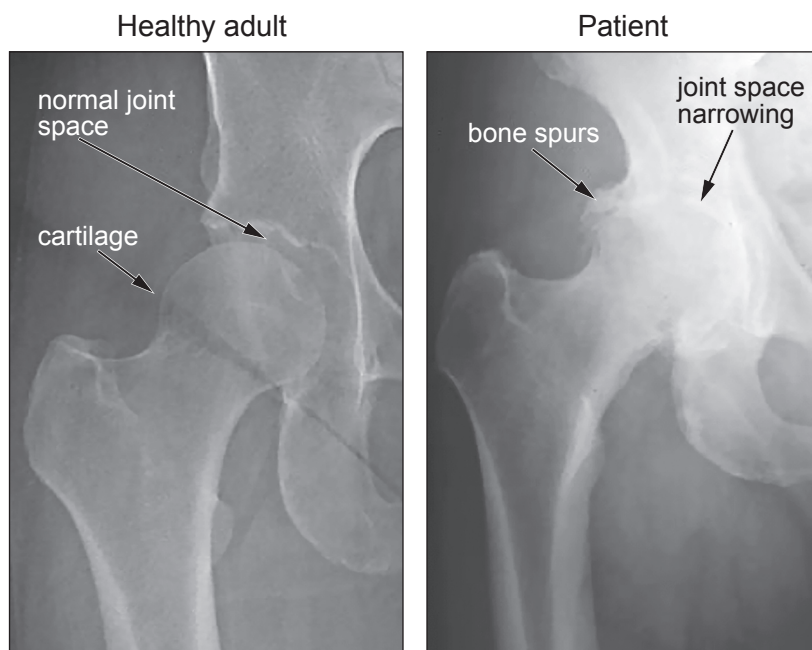
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- (iii) If fractures of the vertebral column are not treated correctly they can lead to problems with posture. Apart from surgery, suggest **one** other type of treatment that could be offered to a patient to avoid posture issues. [1]

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- (c) Two X-ray images of the hip joint are shown below. The X-ray image on the right, shows the hip joint of a patient aged 56 who was overweight. The patient was in a lot of pain and had difficulty walking but had previously played a lot of sport.



- (i) State the type of joint shown. [1]

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- (ii) Using all the information provided, conclude what type of degenerative disease is affecting the patient. Describe the evidence that supports your diagnosis. [2]

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